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SCHOOL OF COMPUTER ENGINEERING (SCE)

Major Project Presentation (Review 2) Sem V/TY B.Tech.

Project Title : "Semiconductor Wafer Defect Detection using Machine Learning"

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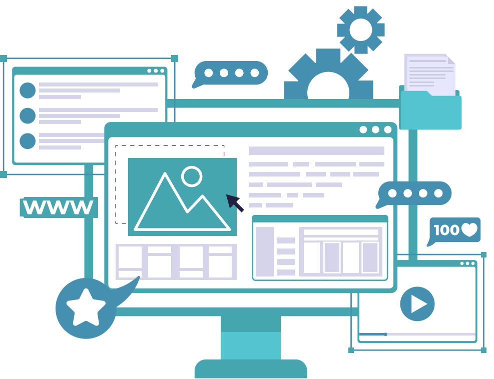


Concept of the project

- This project aims to detect water-related defects in semiconductor wafers using machine learning.
- Traditional methods are often slow and inaccurate.
- By using ML algorithms, we will develop a fast, automated, and highly accurate system for defect detection, enhancing the overall manufacturing process.

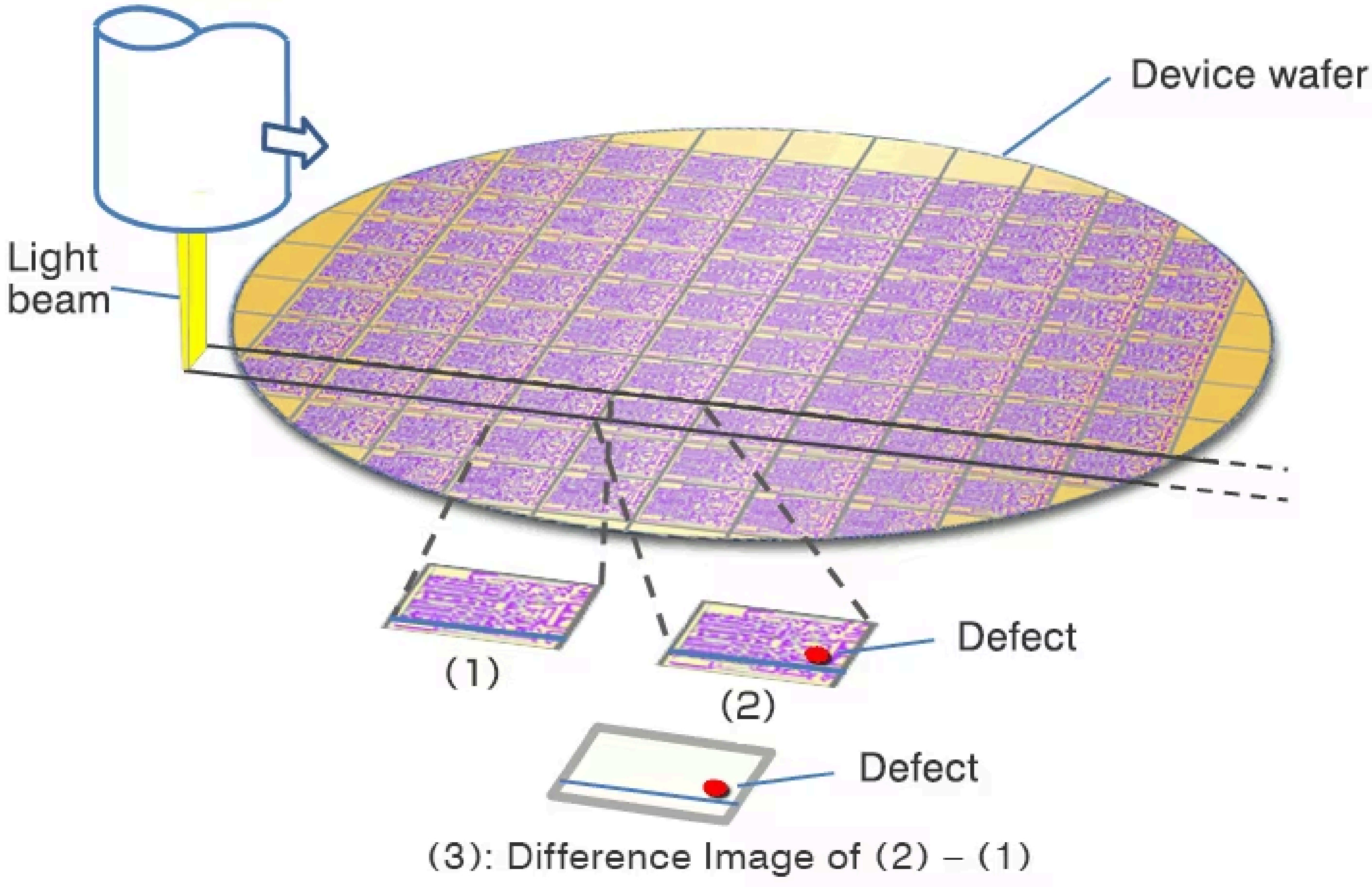
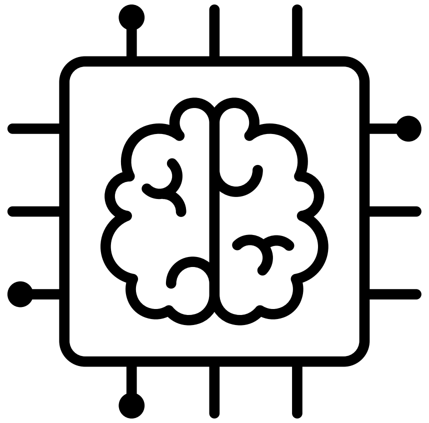


Fig: wafer inspection process used in semiconductor manufacturing

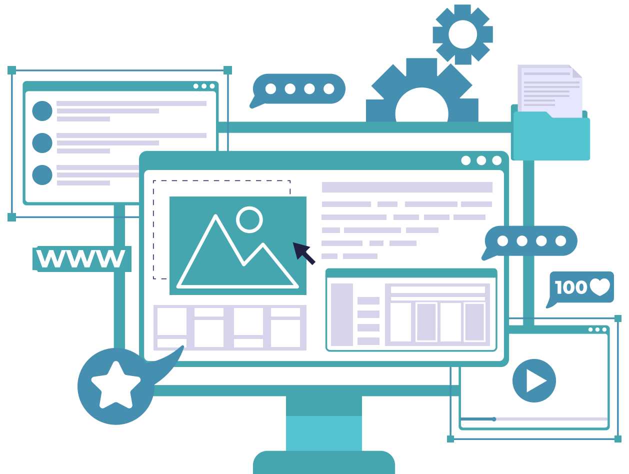
Problem Statement and Objectives

“Semiconductor Wafer Defect Detection using Machine Learning’’.

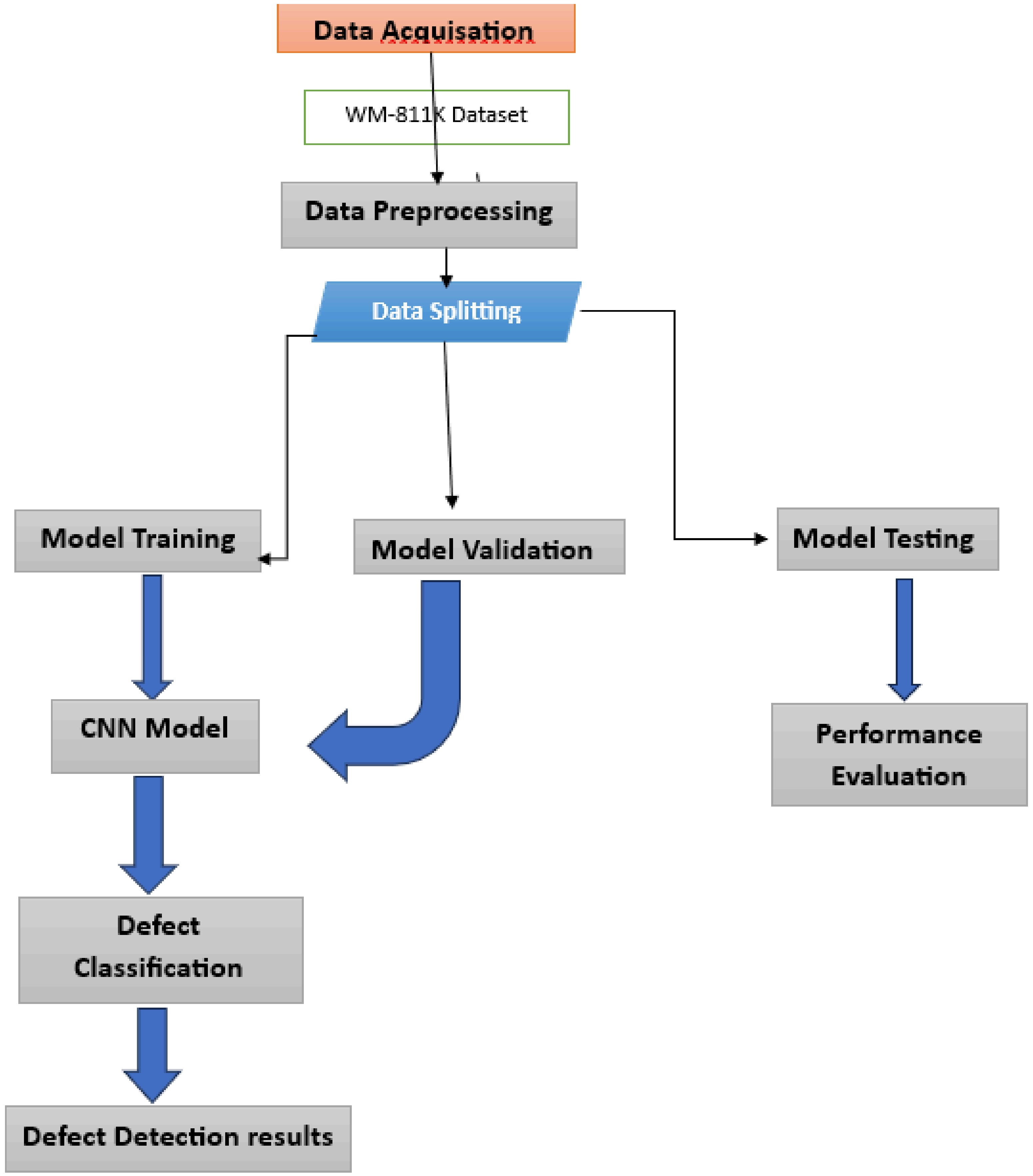
- **Objectives:**
  - To develop a machine learning-based system for detecting defects in semiconductor water processes.
  - To improve the accuracy and efficiency of defect detection.
  - Gather and preprocess semiconductor wafer images to train and test the ML model.



- **Work to be done in this semester :**
    - Data Preparation: Acquire and preprocess the WM-811K dataset.
    - Model Development: Train and fine-tune the model.
    - Validation & Testing: Validate and test the model with split data.
    - Defect Classification: Implement defect detection using the model.
    - Performance Evaluation: Analyze and evaluate model accuracy.



Block diagram and Explanation



Explanation

- **Data Acquisition:** Utilizes the WM-811K dataset containing images of semiconductor wafers.

- **Data Preprocessing:** Image Resizing: Ensures uniform dimensions for all input images.

- **Normalization:** Scales pixel values to the range [0, 1] for improved training efficiency.

- **Data Splitting:**
- Training Set: Used for model training with labeled examples.
- Validation Set: Assesses model performance and tunes hyperparameters.
- Test Set: Evaluates final model performance on unseen data.

- **Model Training:** Implements a Convolutional Neural Network (CNN) to learn features from images through backpropagation.

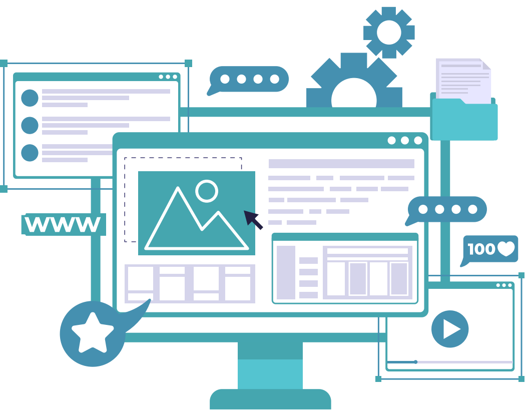
- **Defect Classification:** Classifies images into defect categories based on learned patterns.

- **Performance Evaluation:** Measures accuracy, precision, recall, and F1 score to assess model effectiveness.

Methodology

Approach:

a. **Data Collection:** Gather data from sensors monitoring water quality.
b. **Preprocessing**: Clean and prepare the data for analysis.
c. **Model Development**: Train machine learning models to identify defects.
d. **Testing**: Validate the model against known defects.
e. **Deployment**: Integrate the model into the manufacturing process for real-time detection.



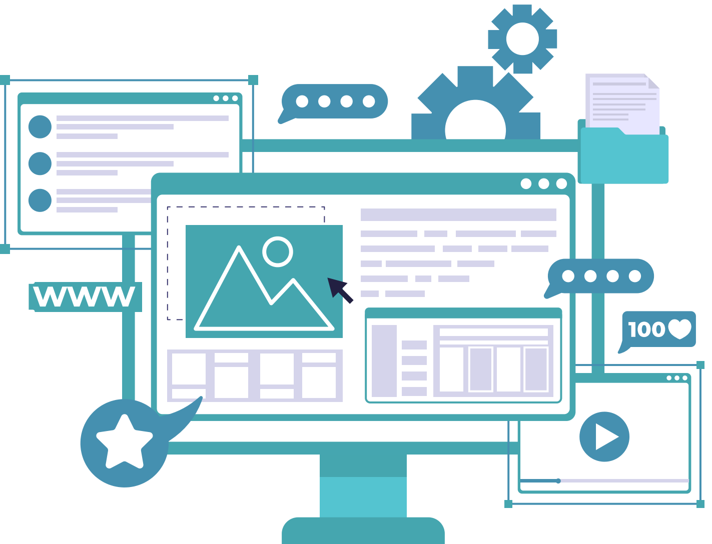
Mathematical understanding

Data Preprocessing:

- Image Resizing:
  $f(x, y) \approx f(0,0)(1-x)(1-y) + f(1,0)x(1-y) + f(0,1)(1-x)y + f(1,1)xy$
- Normalization:
  $x_{\text{normalized}} = \frac{x - \min(x)}{\max(x) - \min(x)}$
  Or standardization:
  $x_{\text{standardized}} = \frac{x - \text{mean}(x)}{\text{std}(x)}$

Convolutional Neural Network (CNN):

- Convolution Operation:
  $(I * K)(i, j) = \sum\sum I(m, n)K(i - m, j - n)$
- Activation Function (ReLU):
  $\text{ReLU}(x) = \max(0, x)$



Model Training:

- Loss Function (Cross-Entropy):

$$L = -\sum y_i \log(p_i)$$

Model Evaluation Metrics:

- Accuracy:

$$\text{Accuracy} = \frac{\text{True Positives}+\text{True Negatives}}{\text{Total Samples}}$$

- Precision:

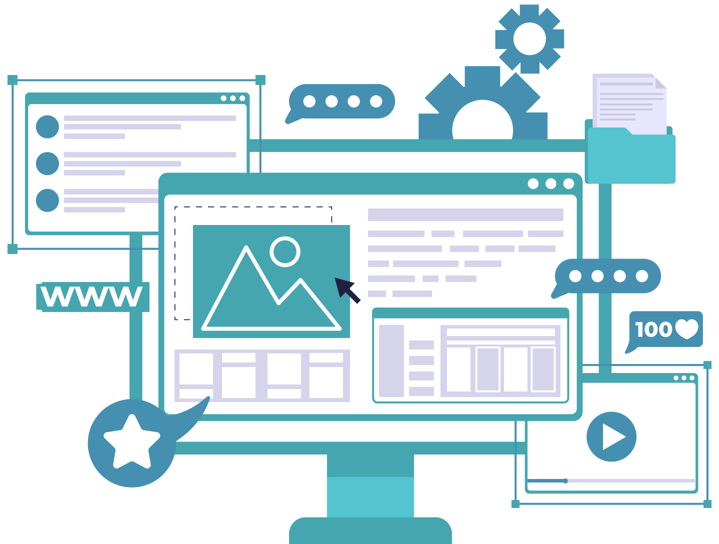
$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives}+\text{False Positives}}$$

- Recall:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives}+\text{False Negatives}}$$

- F1 Score:

$$\text{F1} = 2 \cdot \frac{\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}}$$

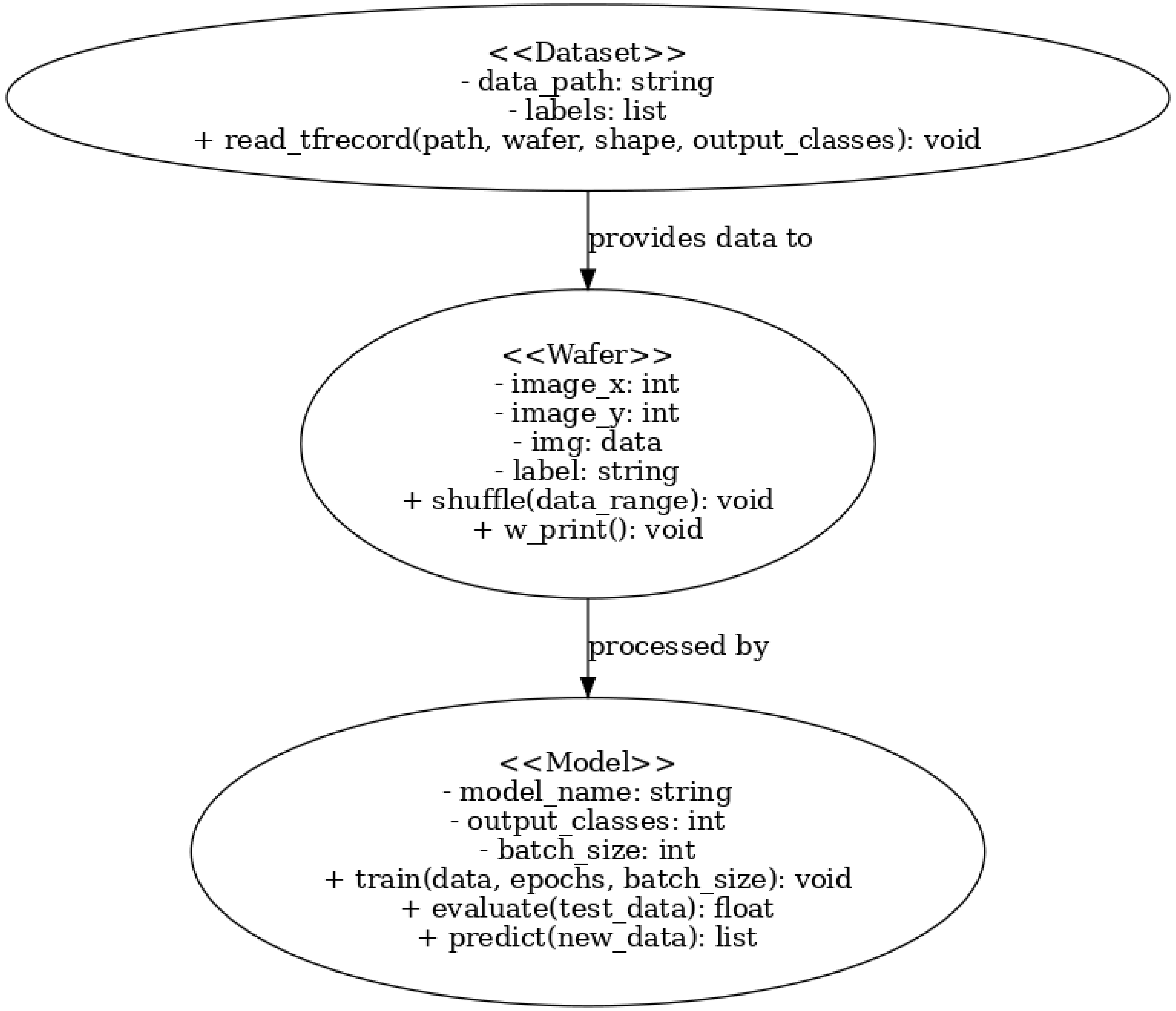


Defect Classification:

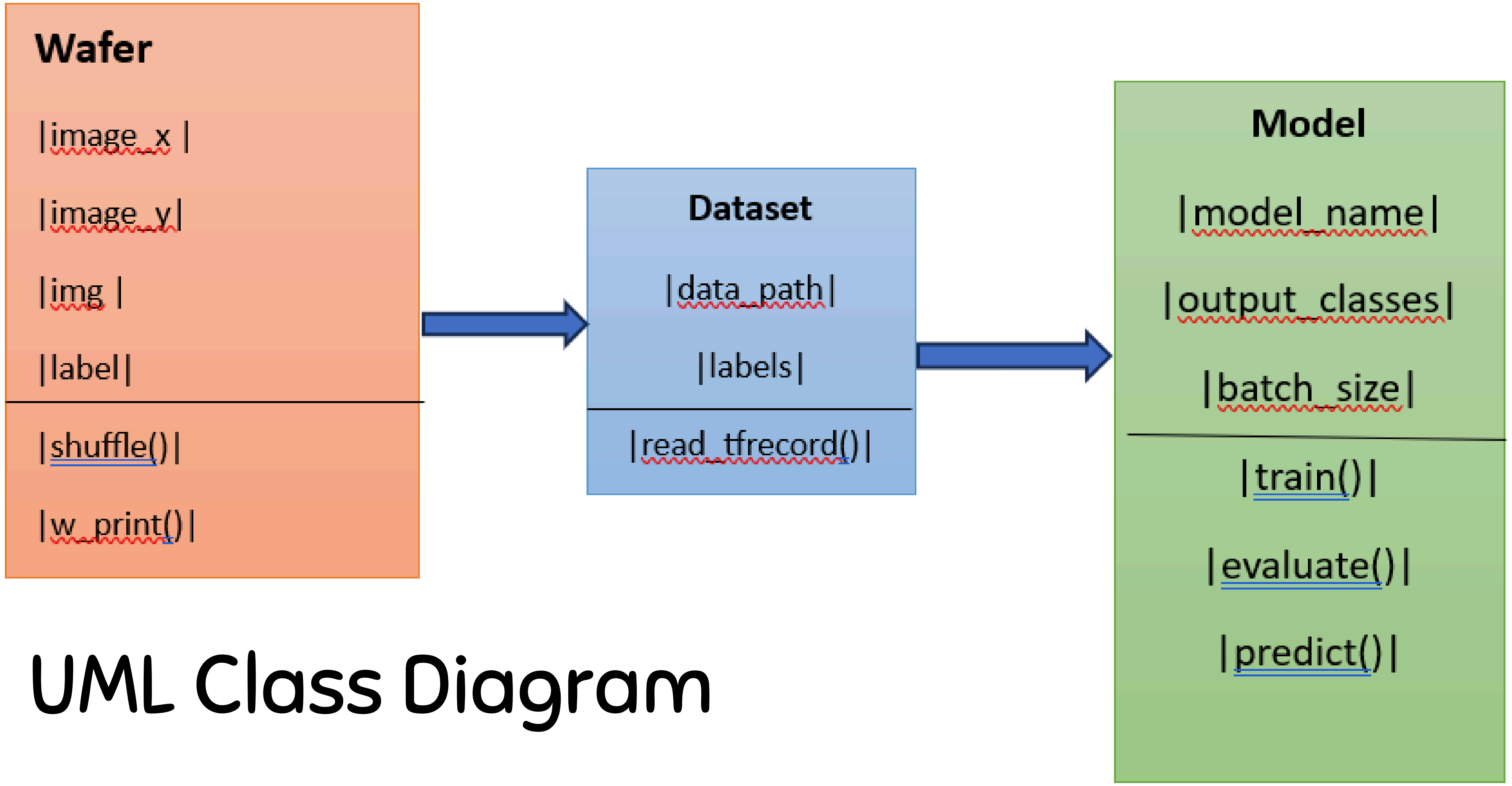
- Class Prediction:

$$\text{predicted\_class} = \text{argmax}(\text{softmax}(z))$$

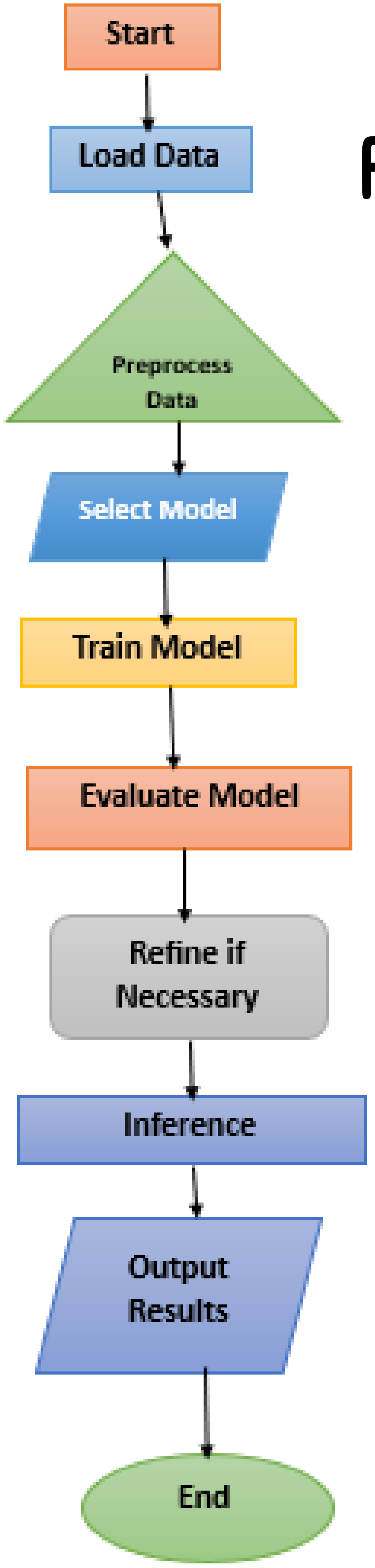
Project design



Visual Representation

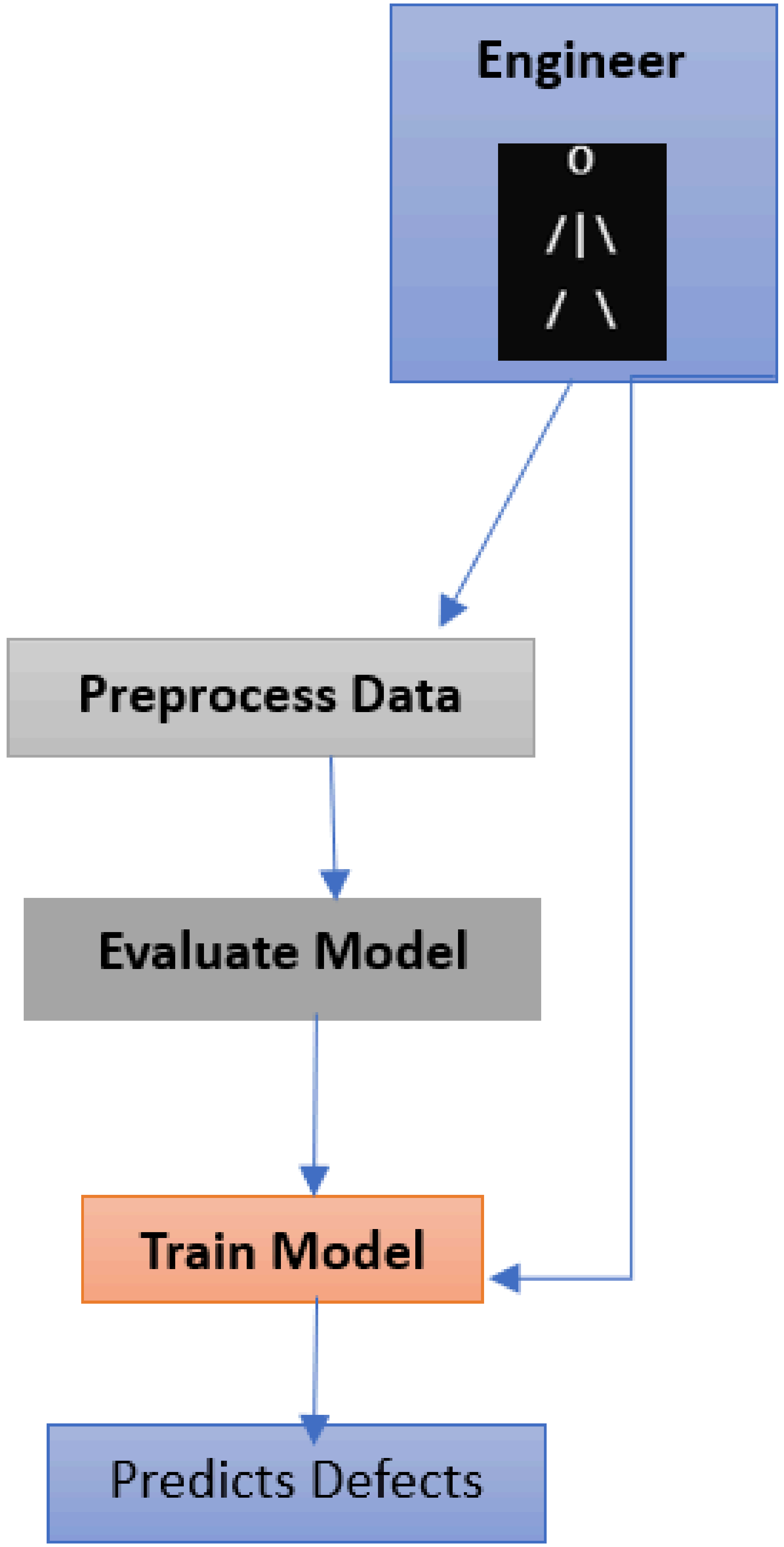


UML Class Diagram

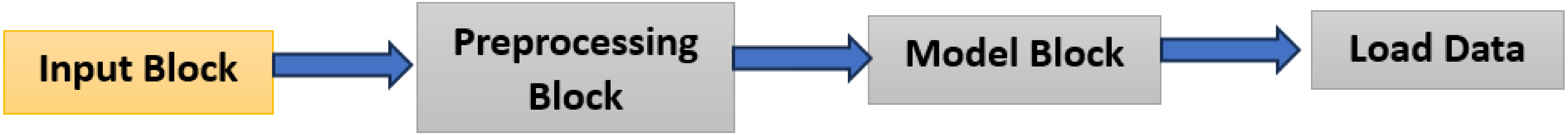


Flowchart

Use case Diagram



Block Diagram



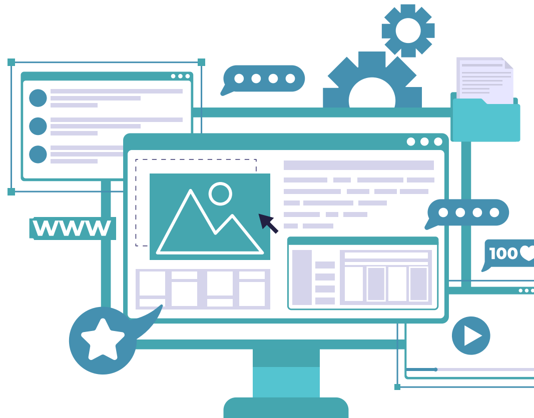
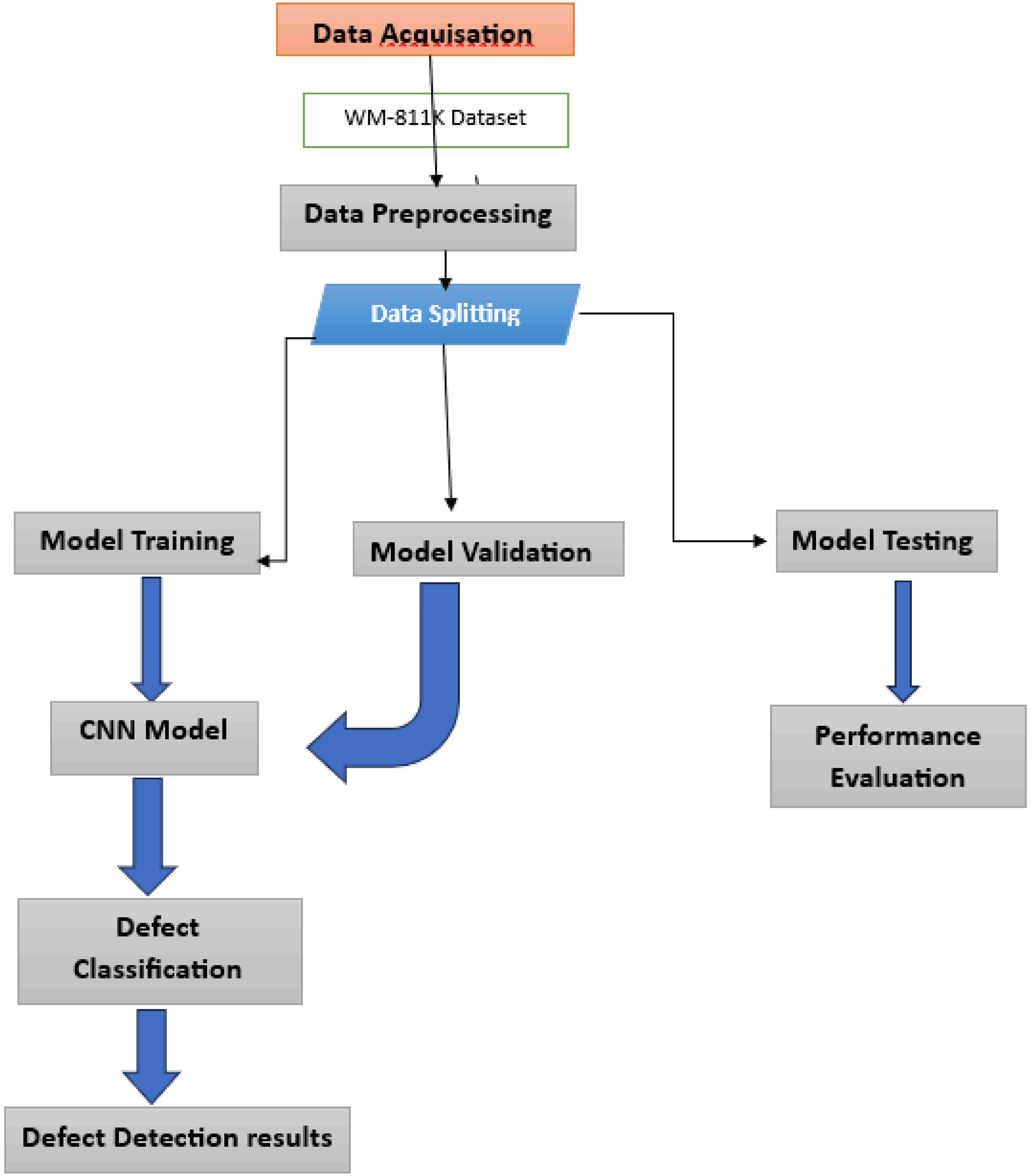


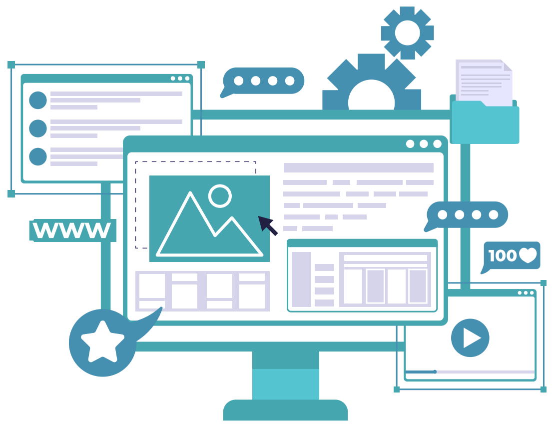
Figure:

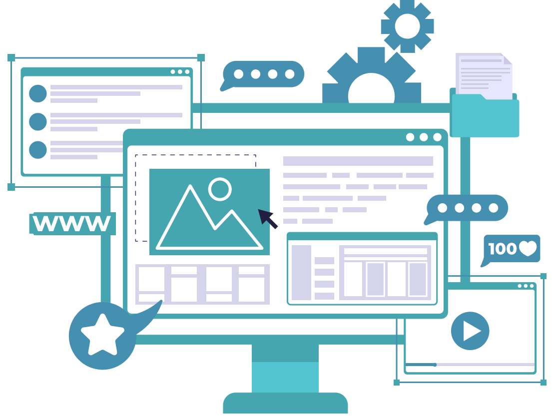

Data Acquisation
WM-811K Dataset
Data Preprocessing
Data Splitting
Model Training
Model Validation
Model Testing
CNN Model
Performance Evaluation
Defect Classification
Defect Detection results

Project Implementation

- **Class: Wafer**
- **Attributes:**
- image_x: Width of the wafer image.
- image_y: Height of the wafer image.
- img: Wafer image data.
- label: Defect label for classification.
- **Methods:**
- shuffle(data_range): Randomizes the dataset order.
- w_print(): Displays wafer properties.

- **Class: Dataset**
- **Attributes:**
    - data_path: Path to the dataset.
    - labels: Defect labels.
- **Methods:**
    - read_tfrecord(path, wafer, shape, output_classes): Reads the dataset and assigns images and labels.



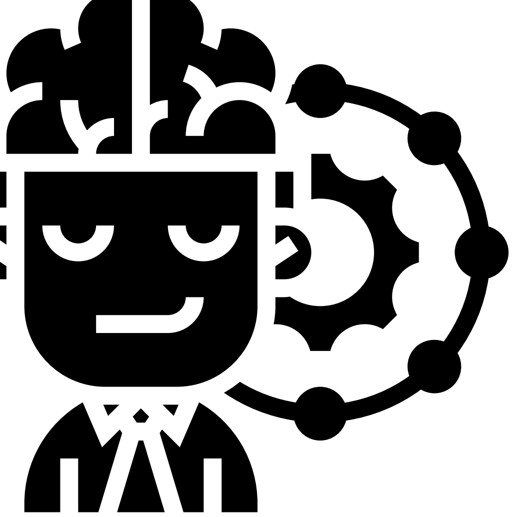


Class: Model

- **Attributes:**
- model_name: Name of the model (e.g., ResNet, Inception).
- output_classes: Number of defect categories.
- batch_size: Size of training batches.
- **Methods:**
- train(data, epochs, batch_size): Trains the deep learning model.
- evaluate(test_data): Validates the model with test data.
- predict(new_data): Predicts defects for new wafer samples.

Relationships:

- Dataset → Model: Supplies data for training, evaluation, and testing.
- Dataset → Wafer: Processes data into usable image and label formats.



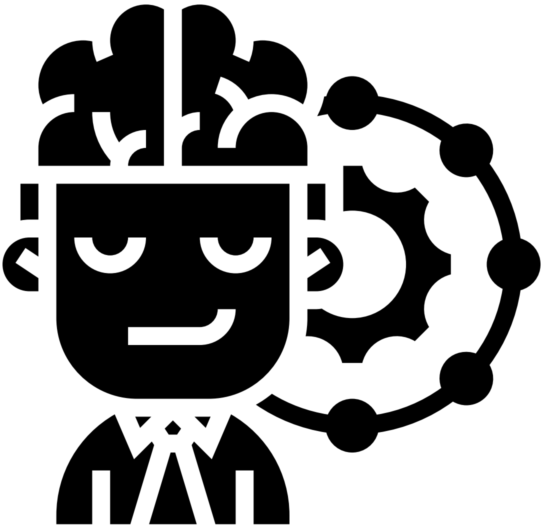
Selected References

- AI-based Inline Inspection Improves Semiconductor Wafer Quality. https://www.intel.com/content/www/us/en/content-details/821377/ai-based-inline-inspection-improves-semiconductor-wafer-quality.html

Citations:

- Include references to key studies, papers, and resources used in the Literature Survey and Methodology sections.
- Follow a consistent citation style (e.g., APA, IEEE).

- https://docs.google.com/spreadsheets/d/1ZJqjmx1dteJLcpHvShLAxOpOR5VXvabD1RzVzWZeEp4/edit?usp=sharing



Thank You !